

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 50

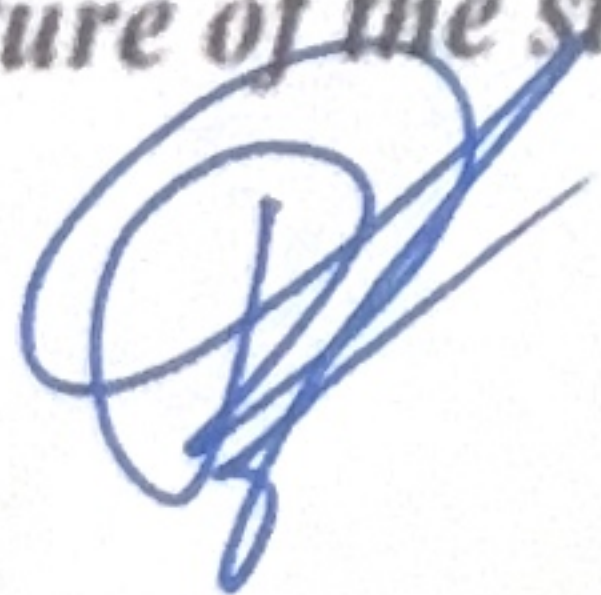
Annexure A

Scenario Based Assignment

Code of Conduct:

I Ravi Teja 192565089 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



19 + 19 + 22 + 18 + 12
= 90
19

Question 1: Constraint Satisfaction Problem (CSP)

1) Formulation as a CSP

- Variables.
- Domains
- Constraints

Variables:

Each arriving flight is a Variable

eg:

F1 = flight 1

F2 = Flight 2

F3 = Flight 3

F4 = Flight 4

The Variable represent gate assigned to each flight

Domain :

Domain contains the gate that can be assigned.

• F1 $\rightarrow$ G1, G2, G3

• F2 $\rightarrow$ G1, G2, G3

• F3 $\rightarrow$ G2, G3

Constraints:

1. No overlapping flights can use the same gate

2. Wide-body aircraft must use suitable wide body gates.

3. Ground crew must be available for the assigned gate and time.

4. Delayed connecting flight should receive suitable gate.

2. Backtracking Search:

1. Assign $F1 \rightarrow G1$

2. Check whether the assignment violate any constraint

3. Assign $F2 \rightarrow G2$

4. Continue assigning gates.

5. if a conflict occurs, undo the previous assignment

6. Try another available gate.

eg:

$F1 \rightarrow G1 \rightarrow F2 \rightarrow G2 \rightarrow F3 \rightarrow G3$

if $F3$ can't use $G3$ because of a constraint, the algorithm goes back and tries another gate.

Constraint Propagation / Forward checking

Forward checking improves the efficiency of Backtracking

For eg, if F_1 is assigned to G_1 , Then another overlapping flight can't use G_1 . $\therefore$, G_1 is removed from the flight's domain

it helps:

- detect conflicts early.
- Reduce the search space.
- Avoid unnecessary Backtracking.

4 Real World challenges and Best Strategy.

- Large number of flights.
- Limited gate
- Flight delays.
- Ground Crew availability.
- Connecting - flight requirement

Conclusion:

CSP With backtracking and forward checking Provides an effective method for solving airport gate assignment problems.

... searching the entire game tree is not Practical.

iii) Alpha - Beta Pruning:

It improves minimax by removing branches can't affect the final decision. The assessment specifically requires efficiency improvement.

Two Values:

- Alpha (α): MAX
- Beta (β): MIN

When: $\alpha \geq \beta$

the remaining branches can be pruned, because they can't change the final decision.

iv) Evaluation Function:

Since searching until the end of the game is expensive a heuristic evaluation function can estimate the quality of board position.

Evaluation = Material advantage + Mobility + Position Advantage.

where:

- Difference in the number / value of piece
- No. of legal moves available
- Control of important board position

v, Recommended strategy.

best approach is :

minimax + Alpha Beta Pruning + Heuristic Evaluation

- Limited Search depth.
- Alpha - Beta ordering
- Good move ordering.
- Fixed time limit for each move.

This provide a balance between good decision-making and fast response time.

Q3 Greedy Search And A^* for Ambulance Routing.

i) Greedy Best-First Search Select the node that appears closest to the destination.

$$f(n) = h(n).$$

where $h(n)$ is estimated distance from the current

location that appears closest to the

accident. it is fast but may not find

the best route.

A* Search:

It uses $f(n) = g(n) + h(n)$

Considering both actual travel cost and estimated distance of a Greedy

Search A* - Ambulance Routing.

iii) Heuristic Accuracy:

Greedy Best-First search
depends strongly on $h(n)$

Accurate heuristic $\rightarrow$ faster and better decision.

Poor heuristic $\rightarrow$ inefficient or risky routes.

iv) effect of Dynamic Road closures:

For eg:

Ambulance $\rightarrow$ Road A $\rightarrow$ Road B $\rightarrow$ Accident

if road B suddenly closes, the original route
cant be followed. They must use Receive the
road-closure update. mark the road as
unavailable.

Recommended Algorithm:

A^* is the best choice for the Ambulance
Routing System. giving better routes under
changing traffic conditions.